Quantile

Einige Quantile der Standardnormalverteilung:

Wkt →
Quantil →

p	$0,9$	$0,95$	$0,975$	$0,99$	$0,995$
N_p	$1,2816$	$1,6449$	$1,96$	$2,3263$	$2,5758$

Einige Quantile der t-Verteilung mit n Freiheitsgraden:

Freiheits-grade →

$n \downarrow p \rightarrow$	$0,9$	$0,95$	$0,975$	$0,99$	$0,995$
1	$3,0777$	$6,3138$	$12,7062$	$31,8205$	$63,6567$
2	$1,8856$	$2,92$	$4,3027$	$6,9646$	$9,9248$
3	$1,6377$	$2,3534$	$3,1824$	$4,5407$	$5,8409$
4	$1,5332$	$2,1318$	$2,7764$	$3,7469$	$4,6041$
5	$1,4759$	$2,015$	$2,5706$	$3,3649$	$4,0321$
6	$1,4398$	$1,9432$	$2,4469$	$3,1427$	$3,7074$
7	$1,4149$	$1,8946$	$2,3646$	$2,998$	$3,4995$
8	$1,3968$	$1,8595$	$2,306$	$2,8965$	$3,3554$
9	$1,383$	$1,8331$	$2,2622$	$2,8214$	$3,2498$
10	$1,3722$	$1,8125$	$2,2281$	$2,7638$	$3,1693$
11	$1,3634$	$1,7959$	$2,201$	$2,7181$	$3,1058$
12	$1,3562$	$1,7823$	$2,1788$	$2,681$	$3,0545$
13	$1,3502$	$1,7709$	$2,1604$	$2,6503$	$3,0123$
14	$1,345$	$1,7613$	$2,1448$	$2,6245$	$2,9768$
15	$1,3406$	$1,7531$	$2,1314$	$2,6025$	$2,9467$
16	$1,3368$	$1,7459$	$2,1199$	$2,5835$	$2,9208$
17	$1,3334$	$1,7396$	$2,1098$	$2,5669$	$2,8982$
18	$1,3304$	$1,7341$	$2,1009$	$2,5524$	$2,8784$
19	$1,3277$	$1,7291$	$2,093$	$2,5395$	$2,8609$
20	$1,3253$	$1,7247$	$2,086$	$2,528$	$2,8453$
21	$1,3232$	$1,7207$	$2,0796$	$2,5176$	$2,8314$
22	$1,3212$	$1,7171$	$2,0739$	$2,5083$	$2,8188$
23	$1,3195$	$1,7139$	$2,0687$	$2,4999$	$2,8073$
24	$1,3178$	$1,7109$	$2,0639$	$2,4922$	$2,7969$
25	$1,3163$	$1,7081$	$2,0595$	$2,4851$	$2,7874$
26	$1,315$	$1,7056$	$2,0555$	$2,4786$	$2,7787$
27	$1,3137$	$1,7033$	$2,0518$	$2,4727$	$2,7707$
28	$1,3125$	$1,7011$	$2,0484$	$2,4671$	$2,7633$
29	$1,3114$	$1,6991$	$2,0452$	$2,462$	$2,7564$
30	$1,3104$	$1,6973$	$2,0423$	$2,4573$	$2,75$

Zufallsvariablen
(Zusammenfassung)

Diskret

Werte: $x_1, x_2, \dots, x_n, \dots$

Wkten: $\mathbb{P}(X = x_i) = p_i$

Wert i

$$\mathbb{P}(a \le X \le b) = \sum_{a \le x_i \le b} p_i$$

Wahrscheinlichkeitsfunktion:

$$\mathbb{P}(X=x) = f(x) = \begin{cases} p_i, & x = x_i \\ 0, & \text{sonst} \end{cases}$$

Verteilungsfunktion:

$$\mathbb{P}(X \le x) = F(x) = \sum_{x_i \le x} f(x_i)$$

Stetig

Werte: Intervalle auf $\mathbb{R}$

Dichtefunktion: $f(x) \ge 0$

Wkten: $\mathbb{P}(X = a) = 0$ ← Punktwahrscheinlichkeit

$$\mathbb{P}(a \le X \le b) = \int_a^b f(x)\,dx$$

Verteilungsfunktion:

$$F(x) = \mathbb{P}(X \le x) = \int_{-\infty}^{x} f(x)\,dx$$

Kennzahlen

Erwartungswert

$$\mu = \mathbb{E}(X) = \sum_{i=1}^{n} x_i \cdot f(x_i)$$

$$\mu = \mathbb{E}(X) = \int_{-\infty}^{+\infty} x \cdot f(x)\,dx$$

Varianz

$$\mathrm{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2$$

$$\sigma^2 = \mathrm{Var}(X) = \underbrace{\sum_{i=1}^{n} x_i^2 \cdot f(x_i)}_{\mathbb{E}(X^2)} - (\mathbb{E}X)^2$$

$$\sigma^2 = \mathrm{Var}(X) = \underbrace{\int_{-\infty}^{+\infty} x^2 \cdot f(x)\,dx}_{\mathbb{E}(X^2)} - (\mathbb{E}X)^2$$

Eigenschaften

$\mathbb{E}(X)$

- $\mathbb{E}(aX+b) = a \cdot \mathbb{E}X + b$
- $\mathbb{E}(X+Y) = \mathbb{E}(X) + \mathbb{E}(Y)$

$\mathrm{Var}(X)$

- $\mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X)$
- $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$

X, Y unabh.

~~Weitere Kennzahlen~~ → Quantile: $\mathbb{P}(X \le Q_p) = p$ ← gegeben

Kovarianz, Korrelation